



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

GENERAL CERTIFICATE OF EDUCATION ADVANCED LEVEL

HORTICULTURE

PAPER 3 Practical Test

6024/3

2 hours 30 minutes

SPECIMEN PAPER

Additional materials:

As listed in Instructions to Supervisors,
Scientific Calculator.

TIME 2 hours 30 minutes + 10 minutes preparation time.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

This paper consists of **three** questions. Answer **all** questions. Marks are given in brackets [] at the end of each question or part question.

You are advised to spend the first **10** minutes carefully reading through the whole paper before answering any question. Spend at least **40** minutes on each question. All writing must be in blue or black ink except for diagrams which must be drawn in pencil. You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE

1	
2	
3	
TOTAL	

This specimen paper consists of 9 printed pages.

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Answer **all** questions on the spaces provided on this question paper. Candidates are advised to desist from using answer booklets.

1. (a) **HS1** and **HS2** are edible parts of horticultural plants:

Carefully cut **HS1** and **HS2** in half using a scapel and draw clearly labelled diagrams showing the internal features of **HS1** and **HS2**.

HS1,

HS2.

[4]

- (b) You will now carry out a food test on **HS1** and **HS2** for the presence of main food types.

chop up one half of **HS1** into small pieces in a petri dish,
grind into a fine paste using the pestle and mortar,
put the ground material in a petri dish,
label the dish **HS1**,

from one half of **HS2**, cut off a **2cm³** piece,
grind to a fine paste using a cleaned pestle and mortar, and place in a clean petri dish,

label the petri dish **HS2**.

- (i) Follow the instructions:

Place a small amount of **HS1** into a dry test tube,
 add **2cm³** of Benedict's solution into the test tube,
 warm the mixture carefully for at least **5** minutes in a water bath,
 record your results and conclusion in the provided table.
 Repeat the procedure with **HS2** and record your findings in the same table.

Sample	Results	Conclusion
HS1		
HS2		

[4]

- (ii) Follow the instructions:

place a small amount of **HS1** onto a white tile,
 add **two** drops of iodine solution using a dropper,
 leave the mixture for about 5 minutes,
 record your results in the table.
 repeat the procedure with **HS2** and record your findings in the same table.

Sample	Results	Conclusion

[4]

(iii) Follow the instructions:

place a small amount of **HS1** into a clean, dry test tube,

add **2cm³** of sodium hydroxide solution and then **two** drops of copper sulphate solution,

leave for **15** minutes while you proceed to the next question,

record your results and conclusion after **15** minutes in the table.

repeat the procedure with **HS2** and record your results in the same table.

Sample	Results	Conclusion
HS1		
HS2		

[4]

(c) Justify the grinding of **HS1** and **HS2** before the tests.

HS3,

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HS4.

.....

[2]

(d) State any **two** precautionary measures taken when carrying out these tests.

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[2]

2. (a) You are provided with **two** weeds **HS3** and **HS4**:

(i) Name the weeds **HS3** and **HS4**.

HS3,

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HS4.

.....

[2]

(ii) Differentiate between **HS3** and **HS4**.

HS3	HS4

[6]

(b) (i) Carefully examine **HS3** and suggest why it survives in dry conditions.

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[1]

(ii) Describe any **two** dispersal methods of **HS3**.

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[2]

- (iii) Carefully examine **HS4** and suggest how it reproduces asexually.

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[1]

- (iv) Explain why it is difficult to control **HS4** amongst smallholder farmers in Zimbabwe.

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[2]

- (c) Describe the effects of weeds on Horticultural plants.

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[4]

- (d) Explain the importance of Integrated Weed Management (**IWM**) to a horticulture farmer.

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[2]

3. (a) You are provided with oven dried soil samples, **HS5** and **HS6** which were collected from two fields, (**compacted field and field which recently received organic matter respectively**), using a core measuring **6cm** diameter and **5cm** long:

- (i) Weigh **HS5** and **HS6** and record the weights.

HS5,

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[1]

- (ii) **HS6.**

.....

[1]

- (b) Calculate:

- (i) volume of core used:

HS5.

[2]

- (ii) bulk density of **HS5** and **HS6**.
HS5,

[2]

HS6.

[2]

- (c) Given that the soil has a particle density of **2,6gcm⁻³**, calculate the porosity of **HS5** and **HS6**.

HS5,

[2]

HS6.

[2]

- (d) Outline any **three** problems associated with **HS5** when used in horticultural production.

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[3]

- (e) Discuss the management practices used by communal and commercial farmers in Zimbabwe to improve the soil structure of **HS5**.

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[5]